

6 (a)	(i) peak voltage = $6\sqrt{2}$ peak voltage = 8.48 V	C1 A1 [2]
	(ii) zero because <i>either</i> no current in circuit (and $V = IR$) <i>or</i> all p.d. across diode	B1 [1]
(b)	waveform: half-wave rectification peak height at about 4.25 cm half-period spacing of 2.0 cm (allow $\pm 1/4$ square for height and half-period)	B1 B1 B1 [3]
(c)	(i) capacitor shown in parallel with resistor	B1 [1]
	(ii) <i>either</i> energy = $\frac{1}{2}CV^2$ or = $\frac{1}{2}QV$ <u>and</u> $Q = CV$ = $\frac{1}{2} \times 180 \times 10^{-6} \times (6\sqrt{2})^2$ = 6.48×10^{-3} J	C1 C1 A1 [3]
	(iii) <i>either</i> fraction = 0.43^2 or final energy = 1.2 mJ fraction = 0.18	C1 A1 [2]
7 (a)	(i) quantum/packet/discrete amount of energy	M1